

SynthScaffold: skip the silent agents

One frozen type-level KV prefix · a different recipe per task
Qwen3-14B · greedy $K=10$

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Verified LatentMAS fork · 0 silent-agent forwards at test time · no 4B claim

The system and the question

Question $\rightarrow$ Planner $\rightarrow$ Critic $\rightarrow$ Refiner $\rightarrow$ KV relay $\rightarrow$ Judger (text)

- Three silent agents, $K=10 \Rightarrow$ **33 extra forwards**, then a Judger.
- Real still beats empty-cache (None). We are not claiming Zou et al. was fake.
- **Question:** can a frozen, question-independent, *type-level* KV prefix replace those silent agents *on the test item*?

This round

Yes on MedQA and GSM8K — **different cache recipes**.

No on AIME: Real is 66.7%; the prefix does not recover it.

Three recipes: how the prefix is built

Same test-time object: one frozen KV table. Silent agents are not run on the test item.

MedQA — Gaussian

- 1 Silent agents on train donors
- 2 Pool $|K|, |V|$ per channel
- 3 Sample i.i.d. Gaussian KV
- 4 Freeze; reuse at test

The model never writes this cache.

GSM8K — Mean-Replay

- 1 Donor *latent steps*, not KV
- 2 Mean by step; restore L2
- 3 Replay on a type prompt
- 4 Model writes KV; freeze

Do not average finished KV tables.

AIME — Medoid

- 1 MATH L5 tapes per subject
- 2 Keep the medoid (no average)
- 3 Replay on a contest prompt
- 4 Assign by last-token cosine

Contest math is not one type.

Main result: two ties, recipes do not transfer

Qwen3-14B, greedy, $K=10$, $T_{\max}=1024$, $B=20$, $n=40$

Task	How the cache is made	Real	Ours	None
MedQA	Gaussian magnitudes	67.5	67.5	32.5
GSM8K	Mean-Replay (model writes KV)	90.0	90.0	80.0
GSM8K	MedQA Gaussian (wrong recipe)	90.0	82.5	80.0
GSM8K	Mean of finished KV tables	90.7	24.7	82.0

- Ours: **0** Planner/Critic/Refiner forwards on the test item.
- Real > None: silent agents do something; we replace that something.
- Mixing recipes is a result, not a footnote: Gaussian $\neq$ Replay.

AIME 2024: Real recovered, medoid does not

Same 30 items, $B=1$, greedy, $T_{\max}=8192$, integer grading (025=25)

Arm	Acc	Cache
Real (silent agents)	20/30 = 66.7%	per-item Planner/Critic/Refiner
Ours (MATH L5 medoid)	17/30 = 56.7%	frozen subject-conditional prefix
None (Judger only)	16/30 = 53.3%	empty

- Real matches Gate-1 / Zou 66.7% (30/30 preds and token counts).
- Medoid is +1 vs None and -3 vs Real (items 4, 10, 18).
- $B=8 \approx 33\%$ was a padded-batch harness bug; ignore it.

Status

AIME stays off the main table. Trusted negative: contest math needs a different object, or the silent agents.

What we will and will not say

Say

- Type-level prefix skips silent agents on MedQA and GSM8K.
- The cache is built differently per type, on purpose.
- Real still beats None; Zou direction holds.
- AIME: Real = 66.7%; Ours $\approx$ None.

Do not say

- One cache for every task.
- 4B result.
- AIME tie, or the old $B=8$ 33%.
- We beat LatentMAS (we match it).
- We skip Judger compute or shrink KV.

Bottom line

Two working recipes, one honest miss. Ready to present as an initial result.